🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.AT.7

Identify Equivalent Algebraic Expressions

Lesson 6-6 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes `Level 1 Support`

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can show that two expressions are equivalent by simplifying and combining like terms.

Language Objective: I can explain my work using the words equivalent, simplify, like terms, and combine.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Equivalent Expressions | Equivalent | Simplify | Like Terms | Combine | Coefficient

👆 *Tap any word to see what it means and a picture.*

1. Expressions with the same value for ____________ allowed number are equivalent.

2. Expressions that always have the same value are ____________.

3. To write an expression in its shortest equal form is to ____________ it.

4. Terms with the same variable raised to the same power, like 4x and 9x, are ____________.

5. To add or subtract like terms into one term is to ____________ them.

6 A number multiplied by a variable is a ______________________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

JUSTIFY

Are these expressions equivalent? How can you prove it?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Equivalent Expressions** — Expressions that look different but have the same value for every allowed variable value.
  Expresiones equivalentes — Expresiones que se ven diferentes pero tienen el mismo valor para cada valor permitido de la variable.

- [ ] **Equivalent** — Expressions that always have the same value.
  Equivalente — Expresiones que siempre tienen el mismo valor.

- [ ] **Simplify** — To write an expression in a shorter, simpler way.
  Simplificar — Escribir una expresión de forma más corta y simple.

- [ ] **Like Terms** — Terms with the same letter raised to the same power, like 2x and 5x.
  Términos semejantes — Términos con la misma letra elevada a la misma potencia, como 2x y 5x.

- [ ] **Combine** — To add or subtract terms with the same letter into one.
  Combinar — Sumar o restar términos con la misma letra en uno solo.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The expressions ____ equivalent because when I simplify 4h + 2h + 30, I get ____. I can prove it by substituting h = ____: both expressions equal ____.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: 3x + 2x combines to 5x, so the two forms are equivalent.

Evidence: Students explain 3x + 2x = 5x (combining like terms), so both equal 5x + 10, and verify by substituting a value like x = 4 to get 30 both ways.

Reasoning: This evidence connects the definition of equivalent to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The expressions ____ equivalent because when I simplify 4h + 2h + 30, I get ____. I can prove it by substituting h = ____: both expressions equal ____.**
- **My answer is ____.**

  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **I know because ____.**

  Lo sé porque ____.
- **So ____.**

  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Equivalent Expressions
2. **Notes 2:** Equivalent
3. **Notes 3:** Simplify
4. **Notes 4:** Like Terms
5. **Notes 5:** Combine
6. **Notes 6:** Coefficient
7. **Watch for this mistake:** *A common mistake in Equivalent Expressions is adding the exponents instead of the coefficients when combining like terms — writing $3x + 2x$ as $5x^2$ instead of $5x$. Remember: $3x$ means 3 groups of x and $2x$ means 2 more groups of x, so you add the coefficients ($3 + 2 = 5$) and keep the variable's power the same (x^1, not x^2).*
8. **Try It 1:** A. 4x — *$3x + x$ is 3 groups of x plus 1 more group of $x = 4x$. Add the coefficients: $3 + 1 = 4$. Check $x = 3$: $3(3) + 3 = 12$ and $4(3) = 12$.*
9. **Try It 2:** A. 2x + 5 — *Order of addition does not change the value (commutative property): $5 + 2x = 2x + 5$. Check $x = 10$: $5 + 2(10) = 25$ and $2(10) + 5 = 25$.*